

Lecture Notes on the Vicsek Model of Collective Motion

1 Motivation

Many natural systems exhibit **collective behavior**:

- Flocking of birds
- Schooling of fish
- Swarming of insects

A key question:

How do simple local rules lead to global order?

The Vicsek model (1995) provides a minimal framework to answer this.

2 Model Description

We consider N self-propelled particles in a 2D box of size $L \times L$ with periodic boundary conditions. Each particle has:

- Position: $\mathbf{x}_i(t)$
- Direction: $\theta_i(t)$
- Velocity: $\mathbf{v}_i = v_0(\cos \theta_i, \sin \theta_i)$

3 Dynamics

3.1 Position Update

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + v_0 \mathbf{e}(\theta_i(t)) \quad (1)$$

3.2 Direction Update

$$\theta_i(t+1) = \text{Arg} \left(\sum_{j \in \mathcal{N}_i} e^{i\theta_j(t)} \right) + \eta \xi_i(t) \quad (2)$$

where:

- $\mathcal{N}_i$: neighbors within radius r
- η : noise strength
- $\xi_i \sim U[-\pi, \pi]$

3.3 Unit Vector

The direction of motion of particle i at time t is represented by a unit vector:

$$\mathbf{e}(\theta_i(t)) = (\cos \theta_i(t), \sin \theta_i(t)) \quad (3)$$

This vector has the following properties:

- It is a **unit vector**, i.e.,

$$\|\mathbf{e}(\theta_i(t))\| = 1$$

- It encodes only the **direction** of motion, not the magnitude
- The actual velocity is:

$$\mathbf{v}_i(t) = v_0 \mathbf{e}(\theta_i(t)) \quad (4)$$

- In 2D, the angle $\theta_i(t)$ fully determines motion

Thus, the Vicsek model separates:

direction (angle) + constant speed

which is key to its simplicity and analytical tractability.

4 Key Ingredients

- **Alignment:** particles align with neighbors
- **Noise:** randomness competes with order
- **Self-propulsion:** constant speed

5 Order Parameter

To quantify collective motion:

$$\Phi = \frac{1}{Nv_0} \left| \sum_{i=1}^N \mathbf{v}_i \right| \quad (5)$$

- $\Phi \approx 0$: disordered phase
- $\Phi \approx 1$: ordered phase

6 Phase Transition

The Vicsek model exhibits a **non-equilibrium phase transition**.

6.1 Control Parameters

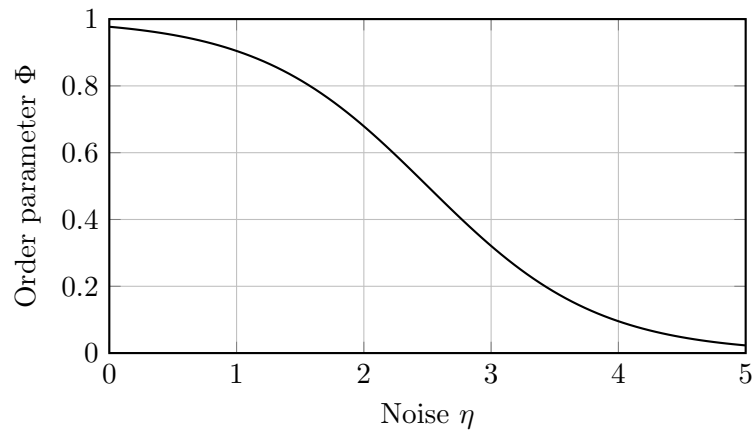
- Noise η
- Density $\rho = N/L^2$

6.2 Behavior

- High noise $\rightarrow$ random motion
- Low noise $\rightarrow$ collective motion

7 Phase Transition Plot

Below is a typical order parameter vs noise plot:



Interpretation:

- Sharp drop indicates phase transition
- Critical noise η_c separates ordered/disordered phases

8 Algorithm

9 Computational Aspects

Naive complexity:

$$\mathcal{O}(N^2)$$

Optimized using:

- Cell lists
- Spatial hashing

Reduced complexity:

$$\mathcal{O}(N)$$

10 Important Phenomena

- Giant density fluctuations
- Traveling bands
- Long-range order in 2D

Algorithm 1 Vicsek Model

```
1: Initialize positions  $\mathbf{x}_i$  and directions  $\theta_i$ 
2: Set parameters  $v_0, r, \eta$ 
3: for  $t = 1$  to  $T$  do
4:   for each particle  $i$  do
5:     Find neighbors  $\mathcal{N}_i$ 
6:     Compute average direction:
```

$$\bar{\theta}_i = \text{atan2} \left(\sum \sin \theta_j, \sum \cos \theta_j \right)$$

```
7:     Update:
```

$$\theta_i \leftarrow \bar{\theta}_i + \eta \xi_i$$

```
8:   end for
9:   for each particle  $i$  do
10:    Update position
11:   end for
12: end for
```

11 Extensions

- Topological interactions
- Inertia-based models
- Coupling with external fields

12 Conclusion

The Vicsek model shows that:

$$\textit{Simple local alignment + noise} \Rightarrow \textit{rich emergent behavior}$$

It remains a foundational model in:

- Statistical physics
- Complex systems
- Network science

Algorithm 2 Vicsek Model

1: Initialize number of particles N , system size L , speed v_0 , interaction radius r , noise η , total time T

2: Initialize positions $\mathbf{x}_i = (x_i, y_i)$ uniformly in $[0, L]$

3: Initialize directions $\theta_i \in [0, 2\pi]$ randomly

4: **for** $t = 1$ to T **do**

5: **for** each particle $i = 1, \dots, N$ **do**

6: Find neighbors:

$$\mathcal{N}_i = \{j \mid \|\mathbf{x}_j - \mathbf{x}_i\| \leq r\}$$

7: Compute average direction:

$$\bar{\theta}_i = \text{atan2} \left(\sum_{j \in \mathcal{N}_i} \sin \theta_j, \sum_{j \in \mathcal{N}_i} \cos \theta_j \right)$$

8: Add noise:

$$\theta_i(t+1) = \bar{\theta}_i + \eta \xi_i(t)$$

9: **end for**

10: **for** each particle $i = 1, \dots, N$ **do**

11: Update position:

$$\mathbf{x}_i(t+1) = \mathbf{x}_i(t) + v_0(\cos \theta_i, \sin \theta_i)$$

12: Apply periodic boundary conditions:

$$\mathbf{x}_i \leftarrow \mathbf{x}_i \bmod L$$

13: **end for**

14: Compute order parameter:

$$\Phi(t) = \frac{1}{N} \left| \sum_{i=1}^N (\cos \theta_i, \sin \theta_i) \right|$$

15: **end for**

Algorithm 3 Vicsek Model with Velocity Order Parameter

1: Initialize N, L, v_0, r, η, T

2: Initialize positions $\mathbf{x}_i$ uniformly in $[0, L]$

3: Initialize directions $\theta_i \in [0, 2\pi]$

4: **for** $t = 1$ to T **do**

5: **for** each particle i **do**

6: Find neighbors:

$$\mathcal{N}_i = \{j \mid \|\mathbf{x}_j - \mathbf{x}_i\| \leq r\}$$

7: Compute average direction:

$$\bar{\theta}_i = \text{atan2} \left(\sum_{j \in \mathcal{N}_i} \sin \theta_j, \sum_{j \in \mathcal{N}_i} \cos \theta_j \right)$$

8: Update direction with noise:

$$\theta_i \leftarrow \bar{\theta}_i + \eta \xi_i$$

9: **end for**

10: **for** each particle i **do**

11: Velocity:

$$\mathbf{v}_i = v_0(\cos \theta_i, \sin \theta_i)$$

12: Update position:

$$\mathbf{x}_i \leftarrow \mathbf{x}_i + \mathbf{v}_i$$

13: Apply periodic boundary:

$$\mathbf{x}_i \leftarrow \mathbf{x}_i \bmod L$$

14: **end for**

15: Compute normalized average velocity:

$$\mathbf{V}(t) = \frac{1}{N} \sum_{i=1}^N \mathbf{v}_i$$

16: Order parameter:

$$\Phi(t) = \frac{|\mathbf{V}(t)|}{v_0}$$

17: **end for**
